

SIMMONS AAF, NC, USA

WMO: 746930

Lat: 35.133N

Lon: 78.933W

Elev: 244

StdP: 14.57

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93737

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.2	25.2	2.7	6.4	28.7	7.9	8.3	32.6	21.7	50.6	19.5	49.6	4.5	340	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	96.1	75.4	93.6	75.2	91.4	74.7	78.7	89.0	77.8	87.6	77.1	86.4	7.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	139.7	81.5	75.4	134.1	80.8	74.6	130.8	80.6	42.3	88.9	41.3	88.0	40.5	86.1	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	15.8	12.9		13.4	100.0	6.5	2.4	8.7	101.8	4.9	103.2	1.2	104.6	-3.6	106.3
				11.7	80.6	5.9	1.5	7.5	81.7	4.0	82.6	0.7	83.5	-3.6	84.6
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.2	43.2	47.6	53.7	63.3	71.4	78.8	81.5	79.8	74.4	64.0	52.5	47.4
	DBStd	15.44	10.28	10.03	9.51	8.01	6.67	5.34	3.84	4.26	5.56	7.64	8.31	9.53
	HDD50	725	262	151	72	5	0	0	0	0	0	2	67	166
	HDD65	2733	678	491	368	126	23	1	0	0	4	112	381	549
	CDD50	5551	50	84	187	405	663	864	977	925	731	435	143	87
	CDD65	2085	1	5	18	76	221	415	512	460	285	80	7	5
	CDH74	20031	5	35	167	710	1967	4241	5426	4522	2275	623	40	20
	CDH80	8387	0	1	22	176	716	1960	2535	1987	843	145	1	1
Wind	WSAvg	5.7	5.9	6.4	6.9	7.0	6.1	5.5	5.3	4.7	5.1	5.1	5.3	5.4
Precipitation	PrecAvg	46.8	3.8	3.5	4.0	3.0	3.3	4.4	5.7	5.4	4.5	3.0	2.9	3.1
	PrecMax	65.8	7.8	6.6	8.9	7.2	10.0	15.9	14.6	10.7	14.7	9.3	6.4	7.7
	PrecMin	32.4	0.8	0.6	0.3	0.2	0.6	0.4	0.6	1.1	0.0	0.5	0.3	0.5
	PrecStd	7.1	1.7	1.7	1.8	1.6	1.8	2.7	2.7	2.5	3.2	2.1	1.7	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.3	78.7	83.4	88.2	95.0	99.3	98.6	98.3	93.5	88.5	78.0	77.0
		MCWB	63.0	64.1	63.1	67.1	72.3	74.9	76.3	76.1	73.2	71.3	64.8	66.2
	2%	DB	68.7	73.3	78.9	84.6	90.3	95.4	95.5	94.9	90.7	83.8	74.5	72.1
		MCWB	60.4	60.6	62.4	66.2	70.7	74.8	76.3	75.8	73.3	69.8	62.7	64.9
	5%	DB	64.4	69.3	74.9	81.6	87.2	92.7	93.2	92.1	88.2	80.6	71.2	68.1
		MCWB	57.5	58.8	60.1	64.4	69.8	74.3	76.2	75.5	73.0	67.9	61.8	63.3
	10%	DB	60.7	64.9	70.5	78.4	84.0	90.0	90.9	89.6	85.5	77.2	67.5	63.7
		MCWB	54.2	56.4	57.8	63.3	68.9	73.5	75.6	75.0	72.1	67.0	60.3	59.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.8	67.1	67.8	72.3	76.2	78.5	80.5	79.9	78.0	76.5	69.4	70.1
		MCDB	70.1	73.5	75.7	79.5	85.8	90.2	91.3	90.8	85.4	80.6	72.1	72.7
	2%	WB	63.3	64.4	65.4	69.9	73.8	77.3	79.1	78.5	76.7	74.3	66.9	67.0
		MCDB	66.8	70.3	73.4	77.6	82.9	89.0	89.3	88.5	84.0	79.4	71.0	70.0
	5%	WB	59.6	61.6	63.5	68.0	72.4	76.3	78.0	77.6	75.4	72.0	64.4	63.4
		MCDB	63.0	67.0	70.2	76.9	81.5	87.4	88.1	86.8	82.7	77.2	69.0	66.4
	10%	WB	54.9	57.8	61.2	66.0	71.1	75.3	77.2	76.8	74.3	69.0	61.3	59.7
		MCDB	58.9	64.3	68.7	75.0	80.0	85.7	86.9	85.4	81.0	74.8	66.6	64.0
Mean Daily Temperature Range	5% DB	MDBR	19.5	21.3	22.3	22.2	20.3	18.9	18.0	17.8	17.8	20.8	21.6	19.3
		MCDBR	24.4	25.4	26.2	25.2	23.1	22.1	20.2	20.5	21.1	22.9	24.7	22.2
		MCWBR	17.2	16.0	12.8	10.4	8.2	6.5	6.1	6.2	7.0	10.4	15.1	15.4
	5% WB	MCDBR	21.2	23.2	21.3	20.5	19.2	19.2	18.0	17.9	17.1	16.4	19.6	19.8
		MCWBR	18.2	16.8	13.4	10.8	7.8	6.8	6.4	6.2	6.6	8.9	15.2	16.0
Clear-Sky Solar Irradiance	taub	0.321	0.334	0.369	0.408	0.464	0.501	0.548	0.524	0.431	0.391	0.341	0.319	
	taud	2.481	2.437	2.374	2.263	2.149	2.088	1.975	2.030	2.290	2.367	2.466	2.522	
	Ebn at Noon	280	288	286	279	263	252	239	243	262	264	269	274	
	Edh at Noon	27	31	36	43	49	51	57	53	39	33	27	24	
All-Sky Solar Radiation	RadAvg	840	1050	1389	1743	1916	2023	1922	1734	1466	1200	922	731	
	RadStd	53	118	102	119	149	137	119	88	121	152	74	61	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (66 neighbors)		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+283	+203

Nomenclature: See separate page